

The Effects of Automation on Population Mobility

Tong Ye

Abstract

This paper studies the effects of automation on the geographic mobility of the working-age population in the U.S. between 1987 to 2020. I exploit cross-commuting zone variation in the exposure to automation and address the endogeneity with a Bartik-style measure. The findings of this study highlight distinct effects of different types of automation on migration. On the whole, automation exposure correlates with a decrease in population growth. While this effect is not significant when using the automation-driven labor share decline as a comprehensive measure encompassing all automation technologies, more significant impacts are observed when examining specific automation types. A one standard deviation increase in exposure to software automation significantly reduces population growth by 0.709 percentage points. Similarly, a one standard deviation increase in machinery automation exposure leads to a significant 0.684 percentage point decrease in population growth.

1 Introduction

Rapid technological progress has been a defining feature of recent decades, with automation, as a prominent technological product, being increasingly and extensively employed across various industries for production and services. According to data from the International Federation of Robotics (IFR), the number of industrial robots per thousand workers in the United States stood at 0.34 in 1993, but steadily rose to 1.68 in 2013. While machines and computers have been instrumental in supporting labor, they have also demonstrated the capacity to replace human workers and reshape the nature of labor input. This has raised concerns regarding potential worker displacement and job losses (e.g., Brynjolfsson and McAfee 2014; Ford 2015). Notably, the declining trend of labor’s share in the U.S. economy over the period 1947 to 2022 (Figure 1) reinforces these concerns. During the same period of substantial growth in industrial robot adoption, from 1993 to 2013, the labor share plummeted from 23.1% to 13.2%, representing a significant reduction of 10 percentage points, or equivalently, 43%.

Numerous studies have investigated the impacts of technology on the labor market, including the effects of technology on wage inequality (Katz and Murphy 1992; Acemoglu and Restrepo 2022), employment polarization (Autor et al. 2003; Goos and Manning 2007; Autor and Dorn 2013; Michaels et al. 2014), and aggregate employment (Autor et al. 2015; Acemoglu and Restrepo 2020). Empirical evidences from these studies have shown that automation causes displacement of labor from jobs and population mobility across occupations. In light of this context, this paper specifically focuses on examining the impacts of technology-induced changes on geographic population mobility. As a distinguishing feature of the U.S. economy, workers’ geographic mobility has long been recognized as a vital mechanism for labor markets to adapt to local economic shocks (Blanchard and Katz 1992). Investigating the impact of automation on the migration patterns of U.S. workers provides a foundation for understanding the broader effects of automation on various aspects of the labor market. Besides, by studying how automation influences workers’ mobility decisions,

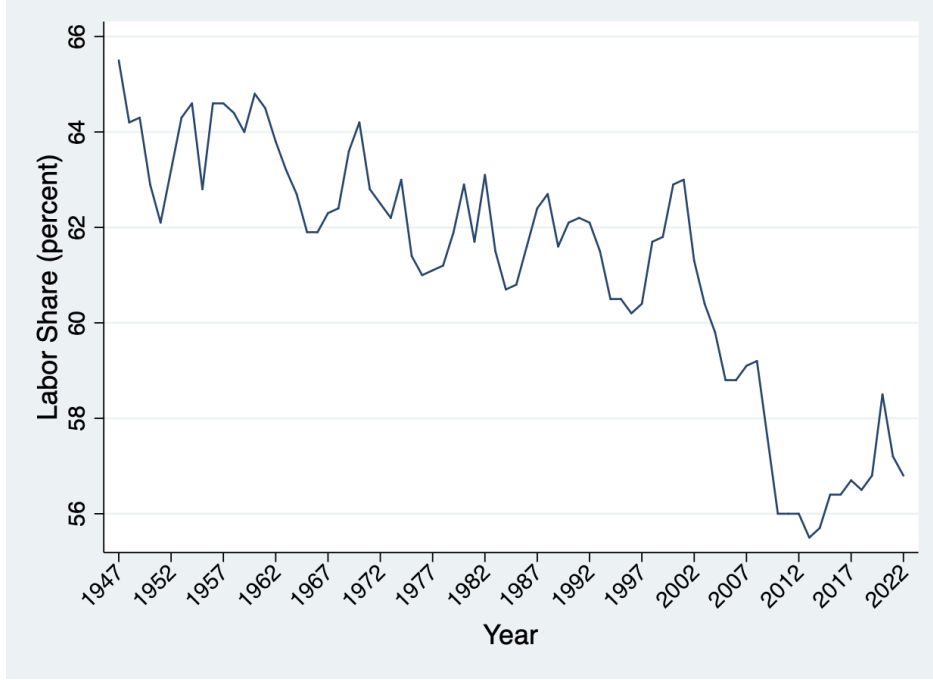


Figure 1: Labor's share of output in the U.S., 1947 to 2022

Note: $labor\ share = \frac{labor\ compensation}{output}$

we can gain valuable insights into the deeper implications of automation on the labor market dynamics, which is essential for policymakers and researchers aiming to formulate effective strategies to address the challenges and opportunities brought by technological advancements to the labor market.

My study investigates the migration effects of automation on the U.S. commuting zone (CZ) level, spanning three time periods, 1987 to 2000, 2000 to 2010, and 2010 to 2020. I focus on the change of working-age population, specifically individuals aged 15 to 64 years, within commuting zones. To identify the causal impact of automation on changes in the working-age population, I employ a two-way fixed effect model that leverages the inter-commuting zone variation in automation exposure.

There are two main challenges for my study: (1) how to measure automation; (2) how to address the endogeneity of CZs' exposure to automation. To gauge the extent of automation, I first use automation-driven decline of the labor share, following the approach of Acemoglu and Restrepo 2022. This approach provides a comprehensive means to capture the scope of

industrial automation. Then, I use two industry-level proxies, change in the value added of dedicated machinery services and change in the value added of specialized software services. These proxies are utilized as distinct measurements to quantify the presence of software automation and machinery automation, respectively. By embracing these methodologies, I aim to offer a comprehensive evaluation of automation’s impact on the population mobility. Data used for measuring automation are drawn from the BLS Total Factor Productivity tables. These tables provide information pertaining to industry labor shares, factor prices, and value added across 61 industries (categorized by 2-digit and 3-digit NAICS codes) spanning the years I studied, from 1987 to 2020.

To address the endogeneity of the exposure to automation, I adopt a Bartik-style measure (Bartik 1991) to quantify each commuting zone’s (CZ) level of automation exposure during a specified period. This measure is constructed based on industry-level automation in the period and baseline industry employment shares of CZs. Data for CZs’ employment shares is sourced from IPMUS Census 5% sample of 1980, which serves as the base year in my study.

The findings of my study reveal distinct effects of various types of automation on migration. Overall, automation exposure is associated with a reduction in population growth. However, this impact is insignificant when utilizing the automation-driven labor share decline as a comprehensive measure encompassing all types of automation technologies. A one standard deviation increase in automation exposure yields a decline in population growth by 0.213 percentage points, lacking statistical significance. Nevertheless, a more pronounced impact becomes evident when delving into the distinct types of automation. One standard deviation increase in exposure to software services significantly reduces population growth by 0.709 percentage points. Similarly, a one standard deviation increase in exposure to machinery services leads to a significant decrease in population growth by 0.684 percentage points.

My work contributes to various branches of literature. First, my study is related to recent works that evaluate workers’ geographic mobility responses to local economic shocks.

Cadena and Kovak leveraged the geographic variation in labor demand during the Great Recession to identify migration responses to local shocks. Their work demonstrated that the location choices of low-skilled Mexican-born immigrants respond strongly to changes in local labor demand (Cadena and Kovak 2016). A recent paper, Greenland et al. 2019, examined the U.S. internal migration response to increased import competition following the granting of Permanent Normal Trade Relations to China in 2001. They found that local labor markets most exposed to the policy change experienced a relative reduction in population growth over the following decade. Additionally, Bartik 2018 paper shows that immobility plays an important role in the earnings difficulties of the non-college educated in the U.S., which indicates the necessity and importance of migration studies in the field of labor economics. My research builds upon these papers by examining workers' mobility in response to the automation shock within the local labor market in the U.S. Consistent with the previous studies, my research firstly shows that working-age population responds to local economic shocks. Secondly, results from my study indicate a negative effect of automation on population growth.

Second, my research complements studies on the labor market impacts of technology discussed earlier. A wealth of studies examine the effects of technology on labor demand, unemployment, and wages. However, there exists a notable gap in the literature concerning the relationship between automation and population mobility, which my study addresses. A closely related study to this paper is the recent work by Faber, Sarto, and Tabellini (Faber et al. 2019). They studied the migration response of American workers to import competition from China and the introduction of industrial robots, demonstrating that a one standard deviation increase in exposure to robots reduces population growth by 0.62 percentage points per decade. Although this study establishes a connection between automation and migration, its focus remains confined to fully autonomous industrial robots, representing only a fraction of the diverse range of automation technologies employed in industrial production and services. In contrast, my research takes a more comprehensive approach by utilizing

macroeconomic indicators. These measures enable me to capture the broader spectrum of automation technologies, along with two distinct types of automation: software automation and machinery automation. This approach allows for a comprehensive evaluation of the effects of automation, thereby providing a more holistic understanding of the relationship between automation and population mobility.

The paper is organized as follows. Section 2 outlines the related literature and the conceptual framework of the study. Sections 3 and 4 detail the empirical strategy and provide an overview of the data used. Section 5 is dedicated to estimating the effects of automation exposure on population mobility. Section 6 undertakes robustness checks to validate the findings, and the paper concludes in Section 7.

2 Conceptual Framework

Automation technology affects labor demand, employment, and wages. The process indicated by empirical evidence from prior research is outlined as follows. Rapid technological progress greatly reduces the cost of accomplishing routine tasks which can be accomplished by machines following explicit programmed rules. As a result, the adoption of technology substitutes workers performing routine tasks. Given that occupations reliant on routine tasks are predominantly situated within the middle range of occupational skills, the demand for workers with middle-level skills experiences a decline. This displacement effect causes an outflow of labor from middle-skilled routine tasks. Furthermore, the technological substitution effect diminishes wages for routine tasks, thereby further contributing to the labor outflow phenomenon.

Conversely, technology, such as computers, complements workers in the execution of non-routine problem-solving and coordination tasks. This innovation fosters the creation of additional service-oriented occupations, notably in fields like customer service. Consequently, the demand for high-skilled professional workers and low-skilled service workers is bolstered. The

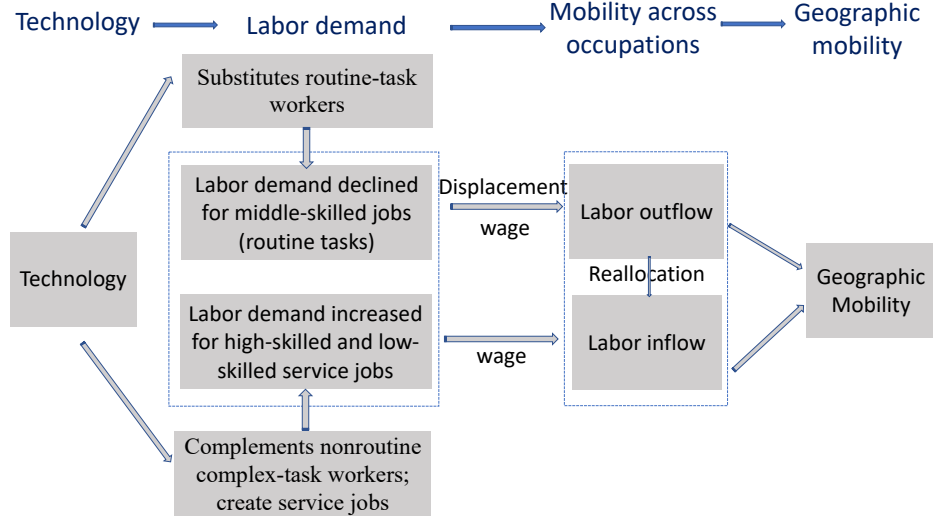


Figure 2: Conceptual Framework

resultant increase in wages for these positions stimulates labor inflow. Moreover, low-skilled workers displaced from routine tasks find their labor redirected toward service occupations. This multifaceted interplay between automation and the labor market intricately shapes the dynamics of labor demand, employment patterns, and wage structures.

Numerous existing studies have shown empirical evidence for this process, Autor, Levy, and Murnane conducted analysis of industry and occupational task input from 1960 to 1998 in the U.S. in their 2003 paper and argued that computerization is associated with reductions of routine manual and cognitive tasks and increases of nonroutine cognitive tasks, which is called “Routinization Hypothesis” (Autor et al. 2003). This argument is supported by Goos and Manning 2007 paper through studying the UK labor market (Goos and Manning 2007). Autor and Dorn 2013 paper conducted a unified analysis of the role of automation in employment and wage polarization in the U.S. between 1980 and 2005. They showed that local labor markets specializing in routine tasks are more likely to adopt computers, experience greater decline in routine employment share, experience employment and wage growth in the low-skilled service sector, and inflows of skilled workers (Autor and Dorn 2013). A recent paper from Acemoglu and Restrepo 2020 estimated effects of robots on employment and wages across commuting zones between 1990 and 2007. The result shows that one more

robot per thousand workers reduces the employment-to-population ratio by 0.2 percentage points and wages by 0.42%, or equivalently, one new robot reduces employment by about 3.3 workers. Besides, another study from them indicates that 50% and 70% of changes in the U.S. wage structure over the last four decades are accounted for by relative wage declines of worker groups specialized in routine tasks in industries experiencing rapid automation (Acemoglu and Restrepo 2022).

Building upon existing research, automation displaces labor from jobs and triggers labor movement across various occupations, driven by shifts in demand and wage structures. This cross-occupation mobility of workers often translates into geographic population mobility. Additionally, as previously discussed, prior studies have demonstrated the responsiveness of workers' location choices to alterations in labor demand resulting from local economic shocks (Cadena and Kovak 2016, Greenland et al. 2019). Given that automation significantly impacts labor demand, it is logical to deduce that it can exert a discernible impact on population mobility. Moreover, the impacts of automation on population dynamics have the potential to extend beyond the mechanisms elucidated in prior studies. Thus, my research serves as an empirical exploration of the relationship between automation and its effects on population mobility.

3 Empirical Strategy

This section introduces the model and measurements used in my study to identify the effects of automation on working-age population mobility. I start by introducing the two-way fixed effect model that constitutes the foundational framework of my analysis. Subsequently, I outline the two primary challenges encountered in my research and elaborate on the strategies employed to effectively address them.

To identify the effect of exposure to automation on internal migration of working-age population across commuting zones in the US, I estimate the following two-way fixed effects

regression model:

$$\Delta \ln Y_{c,t} = \alpha + \beta \text{Exposure to automation}_{c,t} + \delta_c + \gamma_t + \epsilon_{c,t}, \quad (1)$$

where $\Delta \ln Y_{c,t}$ denotes the change in the logarithm of the working-age population (individuals aged 15 to 64 years) living in commuting zone c during period t . My study includes 741 CZs in the US across three time periods between 1987 to 2020, 1987–2000, 2000–2010, and 2010–2020. The treatment variable in my study is the exposure to automation of CZ c in period t , $\text{Exposure to automation}_{c,t}$. δ_c s and γ_t s are the commuting zone fixed effects and the period fixed effects. Given that I estimate a regression in stacked first differences and incorporate both period and commuting zone fixed effects, the coefficients of interest, β , are identified from differences in the exposure to automation between CZs in a given time period adjusting for CZ-specific trends. The underlying assumption of the model is that after conditioning on CZ and period fixed effects, the remaining variation in automation exposure is exogenous.

To identify β as the causal effect of exposure to automation on population mobility, two challenges must be addressed: (1) quantifying automation exposure; (2) addressing the endogeneity of the exposure to automation.

A variety of proxies have been employed in previous literature to measure automation and technology, for example, computerization (e.g., Autor et al. 2003), information and communication technologies (ICT) (e.g., Michaels et al. 2014), and industrial robots (Faber et al. 2019; Acemoglu and Restrepo 2020). While automation can be realized through the utilization of computers, software, and machinery, or often a combination of them, each of the proxies mentioned above solely captures a single aspect of it. Due to this limitation, these proxies are not suitable for comprehensively exploring the impact of automation on migration within the context of my research. The work most closely aligned with my study (Faber et al. 2019) utilizes data on industrial robots sourced from the International Federation of

Robotics (IFR). In accordance with the definition provided by the International Organization for Standardization (ISO), an industrial robot is defined as an “automatically controlled, reprogrammable multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or fixed to a mobile platform for use in automation applications in an industrial environment” (ISO 2021). This strict definition excludes various forms of partial automation technology, such as street sweeper trucks (which require human operators) and mail-sorting machines (that are not reprogrammable). It is worth noting that these partially autonomous machines constitute a substantial portion of industrial machinery and significantly automate a considerable range of tasks, thereby reducing the demand for labor input.

To address the limitations of the measurements used in previous studies, following Acemoglu and Restrepo 2022, I initially measure industry automation during period t with observed declines in industry labor share, denoted as $-\Delta \ln labor\ share_{i,t}$. As there are other determinants of labor shares in addition to automation, such as changes in industry concentration and markups, Chinese import competition, deunionization, i.e. (Elsby et al. 2013; Karabarbounis and Neiman 2014; De Loecker et al. 2020; Autor et al. 2020), I take measures to isolate the automation-driven decline in labor share, represented as $\Delta \ln, \widehat{labor, share}_{i,t}^{auto}$. Specifically, I use two industry-level proxies for adoption of automation technologies: (1) change in the value of dedicated machinery services in value added of the three periods between 1987 and 2020; (2) change in the value of specialized software services in value added of the three periods between 1987 and 2020. I regress changes in industry (log) labor shares of each period on these two proxies for automation technologies, equation (2), and compute the automation-driven decline of labor share as the predicted value in this regression, equation (3):

$$\Delta \ln labor\ share_{i,t} = \alpha_1 + \beta_1 \Delta machinery_{i,t} + \beta_2 \Delta software_{i,t} + \gamma_t + \epsilon'_{i,t}, \quad (2)$$

$$\Delta \ln \widehat{labor\ share}_{i,t}^{auto} = \hat{\alpha}_1 + \hat{\beta}_1 \Delta machinery_{i,t} + \hat{\beta}_2 \Delta software_{i,t} + \hat{\gamma}_t, \quad (3)$$

where γ_t is the period fixed effect. Estimates of β_1 and β_2 from regression equation (2) are 3.21 and 4.51 respectively, both significant under 1% level of significance. Finally, the automation-driven labor share decline is used as the comprehensive measurement of industry automation.

Additionally, I independently leverage the two industry-level proxies, changes in the value of dedicated machinery services in value added and changes in the value of specialized software services in value added. They serve as distinct measurements of software automation and machinery automation within industries. It allows me to individually evaluate the effects of different types of automation on population mobility.

In order to account for potential nonrandom automation exposure of commuting zones (CZs), I address the endogeneity concern by developing a Bartik-style measure (Bartik 1991) to quantify the level of automation exposure for each CZ (c) during each time period (t). To be more precisely, I construct the metric as follows:

$$Exposure\ to\ automation_{c,t} = \sum_{i \in \mathcal{I}} l_{c,i}^{1980} * industry\ automation_{i,t}. \quad (4)$$

$l_{c,i}^{1980}$ represents the baseline employment share of industry i in commuting zone c of the base year 1980, computed as

$$l_{c,i} = \frac{employment\ of\ industry\ i\ in\ commuting\ zone\ c}{total\ employment\ of\ industry\ i}.$$

$industry\ automation_{i,t}$ is measured with different methods discussed earlier in this section. Specifically, when assessing the overall impact of automation, $industry\ automation_{i,t}$ is measured with the automation-driven labor share decline of industry i during period t , $-\Delta \ln labor\ share_{i,t}$. Similarly, for the analysis of software automation and machinery automation, $industry\ automation_{i,t}$ is quantified by the change in the value of dedicated machinery services in value added ($\Delta machinery_{i,t}$) and the change in the value of specialized software services in value added ($\Delta software_{i,t}$), respectively, of industry i during period t .

4 Data and Descriptive Statistics

The main outcome variable in my study, denoted as $\Delta \ln Y_{c,t}$, captures the logarithmic change in the number of individuals falling within the working-age range (15 to 64 years old) residing within commuting zone c in period t . This variable is constructed using population data of US counties, sourced from the intercensal estimates of the US Census Bureau. These estimates provide county-level population data on an annual basis, categorized by age group and other relevant characteristics. To construct this variable, I utilize county-level data for individuals aged 15 to 64 from the years 1987, 2000, 2010, and 2020. Subsequently, I aggregate this data to the commuting zone level for analysis.

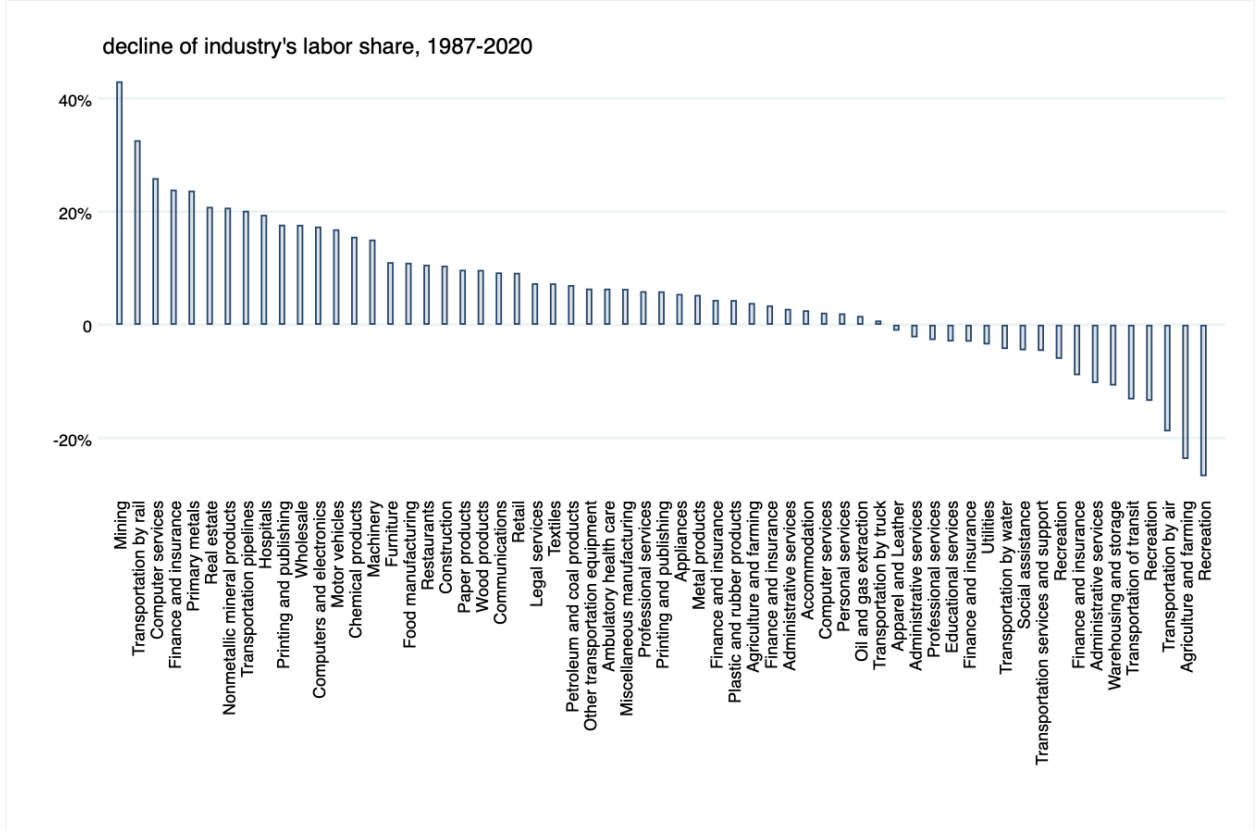


Figure 3: Decline in industry labor shares between 1987 and 2020

Note: The decline showed in the graph is the simple difference of labor shares between 1987 and 2020 of each industry.

I draw on two data sources to construct the exposure to automation variable. First, initial industry employment shares by CZ, $l_{c,i}^{1980}$ s, are obtained from IPUMS U.S. Census 1980 5%

State Sample. The most basic geographic level is the county group, which I used to aggregate the data into CZ level using the crosswalk provided by David Dorn (Autor and Dorn 2013). Second, I use data from the BLS Total Factor Productivity tables on industry labor shares, factor prices, and value added for 61 industries (2-digit and 3-digit NAICS) from 1987 to 2020. The labor share is defined as the portion of labor compensation in the total output. Figure 3 shows the decline of labor share of the industries I studied between 1987 to 2020. The BLS Total Factor Productivity tables also provide the two industry-level proxies for measuring industry software automation and machinery automation, change in the value of dedicated machinery services in value added and change in the value of specialized software services in value added, which are also used in my analysis to predict the automation-driven labor share decline.

The data used for conducting robustness checks in Section 6 is obtained from IPUMS U.S. Census. Population and characteristics of CZs for the year 1980 are sourced from 1980 IMPUS 5% sample. Population data for preceding years are drawn from the IPUMS 1950 1% sample, the 1960 5% sample, and the 1970 1% form 1 metro sample.

5 Results

Table 1 presents the results of the regression model represented by Equation (1). Within the table, the dependent variable corresponds to the change in the logarithmic count of the working-age population during the three distinct periods, 1987–2000, 2000–2010, and 2010–2020, denoted as $\Delta \ln Y_{c,t}$. The independent variable is the commuting zones’ exposure to automation within the specified period, $Exposure_{to, automation_{c,t}}$. This measure is derived via a shift-share approach as delineated in Equation (4), utilizing diverse proxies for industry automation. Notably, all explanatory variables that are displayed in the table are standardized to have a mean of zero and a standard deviation of 1.

The primary findings of my study are presented in columns (2) through (4) of Table 1.

Table 1: Effects of Automation on Working-age Population Mobility

	(1)	(2)	(3)	(4)	(5)
	decline of labor share	auto- driven decline of labor share	change in VA of software	change in VA of machin- ery	change in VA of software and ma- chinery
exposure to automation (labor share decline)	-0.298 (0.170)				
exposure to automation (automation-driven)		-0.213 (0.157)			
exposure to software services			-0.709** (0.226)		-0.463 (0.268)
exposure to machinery services				-0.684** (0.218)	-0.442 (0.259)
constant	9.192* (4.485)	9.146* (4.487)	9.412* (4.475)	9.304* (4.475)	9.424* (4.472)
obs	2,222	2,222	2,222	2,222	2,222

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in the table is the 1987–2000, 2000–2010, and 2010–2020 change in the log count of the working-age population. The independent variable is the exposure to automation, constructed using a shift-share measure with different proxies for industry automation (Equation (4)). Column 1 measures industries’ automation using the changes in the log labor shares. Column 2 uses the predicted change in the log labor share of an industry based on two industry-level proxies for adoption of automation technologies, change in the value of dedicated machinery services in value added and change in the value of specialized software services in value added of the three periods. Column 3, 4 and 5 use the two industry-level proxies directly as measurements of industries’ software automation and machinery automation. All explanatory variables that are displayed are standardized to have a mean of zero and a standard deviation of 1.

Column (2) presents the impact of exposure to automation on the working-age population growth, using the automation-driven labor share decline as a comprehensive measurement of industry automation levels. The result suggests that a one standard deviation increase in exposure to automation leads to a reduction in population growth by 0.213 percentage points per period, although this effect is not statistically significant at the 10% level. In contrast, column (1) gauges the level of automation within industries during a specific period by using the changes in the logarithmic labor shares directly. Under the assumption that the observed

decline in the labor share of an industry, $-\Delta \ln, labor, share_i^{auto}$, can be entirely attributed to automation, the result remains statistically insignificant and of a similar magnitude, at 0.298.

Column (3) and (4) depict the impacts of different types of automation technologies. Column (3) presents the response of working-age population growth to software automation exposure, wherein the industries' software automation are quantified based on changes in the value of specialized software services in value added. The result indicates that a one standard deviation increase in exposure to software automation significantly decreases population growth by 0.709 percentage points per period, under the 1% level of significance. Column (4) shifts to the impact of machinery automation exposure on working-age population growth, evaluating industries' machinery automation through changes in the value of dedicated machinery services in value added. It demonstrates that a one standard deviation increase in exposure to machinery automation significantly decreases population growth by 0.684 percentage points per period, under the 1% level. Column (6) combines both software and machinery automation in the same regression. Despite its lack of statistical significance, the result continues to indicate negative effects.

Overall, my results show that exposure to automation reduces working-age population growth, however, the magnitudes vary when measuring industry automation with different methods. While using the automation-driven decline in labor share as a comprehensive measure covering all types of automation technologies yields an insignificant impact, more pronounced effects emerge when analyzing distinct types of automation, software automation and machinery automation. The presence of these heterogeneous effects underscores the importance of a thoughtful approach to selecting proxies for measuring technology in the relevant research. This choice must align with the specific research context, as it directly shapes the conclusions.

6 Robustness Checks

A concern regarding my analysis is the potential existence of common causal factors behind both the fall in population growth and the rise in the exposure to automation. To verify the exogeneity of the Bartik-measure for exposure to automation in my analysis, I first examine the relationship between commuting zones' characteristics of 1980 and the shares of the top 5 industries based on employment. The majority of the results presented in Table 2 are not statistically significant at the 5% level. The lack of correlation between industry shares and other characteristics of commuting zones suggests that the industry shares of CZs, and consequently the Bartik measure, are exogenous.

Table 2: Relationship between Industry Shares and Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
	retail trade	educational services	hospitals, nursing and resi- dential care facilities	food services and drinking places	crop & animal produc- tion (farms)	pooled
12th grade only	-9.1262 (4.8224)	-8.6972* (4.4094)	-8.4115 (4.4833)	-8.9953 (5.0562)	-10.3797 (5.4533)	-10.6688 (5.5161)
black	0.2264 (1.0603)	0.0716 (1.0373)	0.0711 (1.1518)	0.2323 (1.1051)	0.1408 (1.1511)	0.0223 (1.2009)
veteran	5.4401 (6.8021)	4.4994 (7.4703)	1.6288 (9.7987)	7.0895 (6.5572)	3.4963 (8.8563)	2.5826 (9.0506)
age	0.1553 (0.1083)	0.1335 (0.1161)	0.1812 (0.1420)	0.1218 (0.1051)	0.1973 (0.1315)	0.2153 (0.1381)
male	-17.3587 (10.0984)	-21.2727* (10.7010)	-27.2813* (13.3178)	-17.1285 (9.9499)	-18.5565 (12.4048)	-18.7493 (12.7360)
R^2	0.19	0.20	0.20	0.17	0.20	0.13
N	741	741	741	741	741	39737

Note: Each column reports results of a single regression of a 1980 industry share (in percentage) on 1980 characteristics. The final column is the pooled regression of all industries with standard errors clustered by commuting zone. Results are weighted by 1980 population. Standard errors in parentheses.

Furthermore, I perform a falsification exercise by regressing past changes in population on future changes in automation exposure. Specifically, I regress the changes in working-age

population across commuting zones over the three preceding decades (1950-1960, 1960-1970, and 1970-1980), a period predating significant automation advancements, against the average automation exposure of CZs during the three subsequent periods (1987-2000, 2000-2010, and 2010-2020) that I investigated. The outcomes, presented in Table 3, reveal little evidence

Table 3: Relationship between Future Exposure to Automation and Changes in Working-age Population between 1950 and 1980

	(1)	(2)	(3)	(4)	(5)
	decline of labor share	auto- driven decline of labor share	change in VA of software	change in VA of machin- ery	change in VA of software and ma- chinery
future exposure to automation (labor share decline)	21.989 (48.133)				
future exposure to automation (auto-driven)		9.442 (20.668)			
future exposure to software			3.582 (7.841)		-53.764 (185.182)
future exposure to machinery				6.621 (14.493)	105.993 (350.058)
constant	-1.857 (21.572)	1.315 (15.845)	1.808 (15.048)	1.385 (15.730)	-4.964 (34.909)
obs	2,191	2,191	2,191	2,191	2,191

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in the table is the 1950–1960, 1960-1970, and 1970-1980 changes in the log count of the working-age population. The independent variable is the average exposure to automation during the three periods between 1987 and 2020. All explanatory variables that are displayed are standardized to have a mean of zero and a standard deviation of 1. All regressions include a CZ fixed effect and a period dummy.

suggesting endogeneity or reverse causality. While it does not fully rule out the possibility that other factors contribute to the contemporaneous CZ-level relationship between rising automation exposure and declining population growth during the three periods I studied, estimates in Table 3 demonstrate that this relationship was absent in the pre-period. These results also suggest that the negative effects of exposure to automation on working-age

population growth depicted in Table 1 are not driven by secular trends adversely affecting specific CZs.

7 Conclusion

In this paper, I study the migration effects of automation on the working-age populations in the U.S. To establish the causal impact of automation on population change, I leverage the variation in commuting zones' exposure to automation between 1987 and 2020, utilizing a two-way fixed effect model. To provide a comprehensive evaluation, I first comprehensively capture the scope of Industrial automation using automation-driven decline of the labor share, following the approach of Acemoglu and Restrepo 2022. Then, I use two industry-level proxies, change in the value added of dedicated machinery services and change in the value added of specialized software services, to quantify the presence of software automation and machinery automation, respectively. I address the endogeneity of the exposure to automation through adopting a Bartik-style measure (Bartik 1991) that quantifies each commuting zone's level of automation exposure during a specified period. This measure is constructed based on industry-level automation during the period and baseline industry employment shares in each CZ. Population data used in my study are drawn from the intercensal estimates of the US Census Bureau. Data used to construct CZs' automation exposure is sourced from the IPMUS US Census and the BLS Total Factor Productivity tables.

The findings of my study reveal distinct effects of various types of automation on migration. Overall, automation exposure is associated with a decrease in population growth. However, this impact is insignificant when utilizing the automation-driven labor share decline as a comprehensive measure encompassing all types of automation technologies. A one standard deviation increase in automation exposure yields a decline in population growth by 0.213 percentage points, lacking statistical significance. Nevertheless, a more pronounced impact becomes evident when examining specific automation types. One standard deviation

increase in exposure to software services significantly reduces population growth by 0.709 percentage points. Similarly, a one standard deviation increase in exposure to machinery services leads to a significant decrease in population growth by 0.684 percentage points.

This study adds to the body of research that examines workers' geographic mobility in response to local economic shocks. It also complements existing studies that investigate the influence of technology on the labor market. Moreover, the presence of the heterogeneous effects when measuring technology using distinct methods underscores the importance of selecting appropriate proxies for measuring technology in the relevant research.

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